Fuzzy Sliding-Mode Underactuated Control for Autonomous Dynamic Balance of an Electrical Bicycle

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***Abstract*—The purpose of this paper is to stabilize the running motion of an electrical bicycle. In order to do so, two strategies are employed in this paper. One is to control the bike's center of gravity (CG), and the other is to control the angle of the bike's steering handle. As in general, the control of the CG applies a pendulum. An additional factor is the lean angle with respect to the gravitational direction of the bicycle in motion. In this total, the proposed system produces three outputs that will affect the dynamic balance of an electrical bicycle: the bike's pendulum angle, lean angle, and steering angle. Based on the data of input–output, two scaling factors are first employed to normalize the sliding surface and its derivative. According to the concept of the if–then rule, an appropriate rule table for the ith subsystem is obtained. Then, the output scaling factor based on Lyapunov stability is determined. The purpose of using the proposed fuzzy sliding-mode underactuated control (FSMUAC) is to deal with the huge uncertainties of a bicycle system often caused by different ground conditions and gusts of wind. Finally, the simulations for the electrical bicycle in motion under ordinary PID control, modified proportional–derivative control, and FSMUAC are compared to judge the efficiency of our proposed control method.**

***Index Terms*—Dynamic balance, electrical bicycle, fuzzy sliding-mode underactuated control (FSMUAC), Lyapunov stability, modified proportional–derivative control (MPDC).**

I. Introduction

As ONE KNOWS, a fuzzy control algorithm [1]–[4] consists of a set of heuristic decision rules and is regarded as a nonmathematical control algorithm, which has been proved to be very attractive whenever the controlled systems cannot be well defined or modeled. However, it needs trial and error to obtain an acceptable tracking performance. Aside from the heuristics-based fuzzy control, a model-based fuzzy control (e.g., Takagi and Sugeno model-based control) is another approach to improving the performance of a fuzzy system [5], [6]. In this paper, the combination of sliding-mode control and fuzzy control, which is the so-called fuzzy sliding-mode control (FSMC), provides a robust controller for the nonlinear systems [7]–[9]. Moreover, this paper introduces an FSMC theme with the number of outputs larger than that of the control inputs, something the authors of this paper realized no paper has discussed before. This specific type of FSMC is the so-called fuzzy sliding-mode underactuated control (FSMUAC).

Some papers have discussed underactuated control. An underactuated control was applied to the well-known swing up and balancing control problem of the inverted pendulum [10]. Begovich *et al.* [11] presented a Takagi–Sugeno fuzzy scheme for the real-time trajectory tracking of an underactuated robot: the Pendubot. Lefeber *et al.* [12] addressed the tracking problem for an underactuated ship using surge force and yaw moment. In addition, a modification of the Newton algorithm applied to nonholonomic motion planning with energy optimization was developed by Duleba and Sasiadek [13]. Siqueira and Terra [14] developed the procedure based on the H_∞ technique to obtain the position control of underactuated manipulators. Tarn *et al.* [15] summarized recent theoretical output regulation for nonlinear systems with an experimental study of an underactuated mechanical system: the Pendubot. Recently, the problem of controlling the planar position and orientation of an underactuated surface vessel was addressed by Ghommam *et al.* [16]. In summary, the unstable feature of an underactuated control system (e.g., Pendubot and electrical bicycle) makes the corresponding control problem more difficult.

Over the years, there have been many papers on the topic of the stabilization control of two-wheeled vehicles. In 1971, Sharp [17] analyzed the stability of a straight-running two-wheeled vehicle. He simplified the vehicle into a model of 4 DOF. His work indicated that the steering angle of the bicycle's handle, the side slip of the bicycle's body, the angle of the bicycle's body, and the lean angle between the bicycle's body and the gravitational force are factors that affect the stability of the straight-running two-wheeled vehicle. Subjects regarding the hardware of the two-wheeled vehicle system, including the bicycle's body and tires, were also discussed using this model. Another paper proposed by Grasser *et al.* [18] introduced a novel control method for two-wheeled vehicles that operates by shifting the driver's weight. Then, there are the two papers from Tanaka and Murakami [19] and Niki and Murakami [20], respectively, which realized stable bicycle running through controlling the steering angle of a bicycle's handle. Moreover, recently, the stability of

Manuscript received December 09, 2007; revised June 16, 2008. Manuscript received in final form August 05, 2008. First published April 14, 2009; current version published April 24, 2009. The work of C.-L. Hwang was supported by the National Council of Science of R.O.C. under Grant NSC-96-2221-E-032-012-MY2. The work of H.-M. Wu and C.-L. Shih was supported by the National Council of Science of China under Grant NSC-95-2212-E-011-047. Recommended by Associate Editor C.-Y. Su.

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Digital Object Identifier 10.1109/TCST.2008.2004349

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the bicycle at low speed was studied by Iuchi *et al.* [21]. Summing up the conclusion reached by previous works, the control of a bicycle's center of gravity (CG) and steering-handle angle are two of the most important issues in realizing a stable running motion. When humans ride bicycles, both of these aspects are elegantly accomplished by body control and thus achieving stable bicycle motion. As in general, the control of the CG applies a pendulum. An additional factor is the lean angle with respect to gravitational direction of the bicycle in motion. Overall, the electrical bicycle system has two sets of control inputs: one to control manipulate the pendulum and the other the steering handle; on the other hand, the proposed system generates three outputs that will affect the dynamical balance of the electrical bicycle: the bicycle's pendulum angle, steering angle, and lean angle.

At this point, most previous works employ linear controllers. The paper by Ichui *et al.* [21] merely implemented a linear controller based on a linearized model for his electrical bicycle to avoid instabilities. Astrom *et al.* [22] discussed the dynamics and control of bicycles via a linearized approach as well. These control methodologies only force the operating point into the vicinity of the desired posture or drive the operating point to the target near the desired attitude. In these situations, it is difficult for an electrical bicycle to obtain autonomous dynamic balance when it is encountered with huge uncertainties, e.g., varying ground conditions and external disturbances in the air. A better choice is nonlinear control. In the past, a nonlinear control theme for an underactuated acrobot has been constructed by Berkemeier and Fearing [23]. In fact, for the trajectory of 1 Hz, no tracking error was committed by their controller, as opposed to the tracking error of 13 ° by the pseudolinearizing controller. Their work revealed that the response of a nonlinear dynamic system to a nonlinear controller was much better than that to a linear controller. Recently, a paper developed by Yamaguchi *et al.* using acceleration-based backstepping control [24] confirmed that a nonlinear control for an electrical bicycle is better than that of conventional control. In summary, the purpose of using the FSMUAC method to control the electrical bicycle is to deal with exactly what linear control themes could not: the often huge uncertainties caused by varying ground conditions (e.g., wet ground, muddy ground, and concrete ground) and unexpected gusts of wind.

In the sequel, the proposed FSMUAC is introduced. Due to the underactuated feature of an electrical bicycle, two novel reference signals are designed such that the uncontrolled state (i.e., the lean angle) is simultaneously included into these two control modes (i.e., pendulum and steering angles). Hence, it is indirectly manipulated by these two controllable states. In addition, the theorems about bounded-input–bounded-output and asymptotical tracking for the sliding surface are discussed. Based on the practical ranges of the sliding surface and its derivative, two normalizing scaling factors are selected. The fuzzy rule table of the ith subsystem using sliding surface and its derivative (i.e., the set of if–then rules) is achieved as shown in Table I. Then, the quantity of this fuzzy table is assigned as the value between -1 and 1 (see Table II). This table is skew symmetric about the diagonal line; its absolute value is proportional to the distance from the diagonal line. Then, the crisp control input is determined from Table II by multiplying the output scaling factor, which is chosen according to the Lyapunov stability of the closed-loop system. Generally speaking, a larger output scaling factor is assigned, and a smaller tracking error and a faster response are obtained; however, the risk of transient or oscillatory response after transience occurs. Then, a saturation of control input will result in a possible instability or the excitation of unmodeled dynamics will occur.

TABLE I
RULE TABLE OF THE ith FSMUAC

$\dot{\bar{s}}_i$ / $\bar{s}_i$	PH	PB	PM	PS	PI	ZE	NI	NS	NM	NB	NH
NH	ZE	NI	NS	NM	NM	NB	NB	NB	NH	NH	NH
NB	PI	ZE	NI	NS	NM	NM	NB	NB	NB	NH	NH
NM	PS	PI	ZE	NI	NS	NM	NM	NB	NB	NB	NH
NS	PM	PS	PI	ZE	NI	NS	NM	NM	NB	NB	NB
NI	PM	PM	PS	PI	ZE	NI	NS	NM	NM	NB	NB
ZE	PB	PM	PM	PS	PI	ZE	NI	NS	NM	NM	NB
PI	PB	PB	PM	PM	PS	PI	ZE	NI	NS	NM	NM
PS	PB	PB	PB	PM	PM	PS	PI	ZE	NI	NS	NM
PM	PH	PB	PB	PB	PM	PM	PS	PI	ZE	NI	NS
PB	PH	PH	PB	PB	PB	PM	PM	PS	PI	ZE	NI
PH	PH	PH	PH	PB	PB	PB	PM	PM	PS	PI	ZE

TABLE II
LOOKUP TABLE OF THE ith FSMUAC

$\dot{\bar{s}}_i$ / $\bar{s}_i$	1.0	0.8	0.6	0.5	0.2	0	-0.2	-0.5	-0.6	-0.8	-1.0
-1.0	0.0	-0.05	-0.2	-0.5	-0.7	-0.8	-0.9	-0.95	-1.0	-1.0	-1.0
-0.8	0.05	0.0	-0.05	-0.2	-0.5	-0.7	-0.8	-0.9	-0.95	-1.0	-1.0
-0.6	0.2	0.05	0.0	-0.05	-0.2	-0.5	-0.7	-0.8	-0.9	-0.95	-1.0
-0.5	0.5	0.2	0.05	0.0	-0.05	-0.2	-0.5	-0.7	-0.8	-0.9	-0.95
-0.2	0.7	0.5	0.2	0.05	0.0	-0.05	-0.2	-0.5	-0.7	-0.8	-0.9
0	0.8	0.7	0.5	0.2	0.05	0.0	-0.05	-0.2	-0.5	-0.7	-0.8
0.2	0.9	0.8	0.7	0.5	0.2	0.05	0.0	-0.05	-0.2	-0.5	-0.7
0.5	0.95	0.9	0.8	0.7	0.5	0.2	0.05	0.0	-0.05	-0.2	-0.5
0.6	1.0	0.95	0.9	087	0.7	0.5	0.2	0.05	0.0	-0.05	-0.2
0.8	1.0	1.0	0.95	0.9	0.8	0.7	0.5	0.2	0.05	0.0	-0.05
1.0	1.0	1.0	1.0	0.95	0.9	0.8	0.7	0.5	0.2	0.05	0.0

The organization of this paper is described as follows. In the next section, the dynamics of an electrical bicycle is derived, and the problem statement is also given. In Section III, the properties of underactuated control are given. In Section IV, the controller design of the FSMUAC is discussed, and the stability of the closed-loop system is also verified by Lyapunov stability theory. Then, the compared simulations, including the ideal case, in the face of nonlinear time-varying uncertainty, in the presence of the pulse disturbance to represent a gust of wind, variable speed for the motion of bicycle, and different control cycle time, among ordinary PID control, the MPDC, and the FSMUAC are given in Section V. Finally, the conclusions are arranged in Section VI.

II. MODELING AND PROBLEM FORMULATION

A. Modeling of Electrical Bicycle

In the beginning, the mechanism of an electrical bicycle is shown in Fig. 1. The dynamics of the corresponding electrical bicycle is derived in Appendix A. Based on these results, the following states are defined: $x_1(t) = \theta(t)$, $x_2(t) = \phi_1(t)$, $x_3(t) = \phi_2(t)$, $x_4(t) = \dot{\theta}(t)$, $x_5(t) = \dot{\phi}_1(t)$, and $x_6(t) =$

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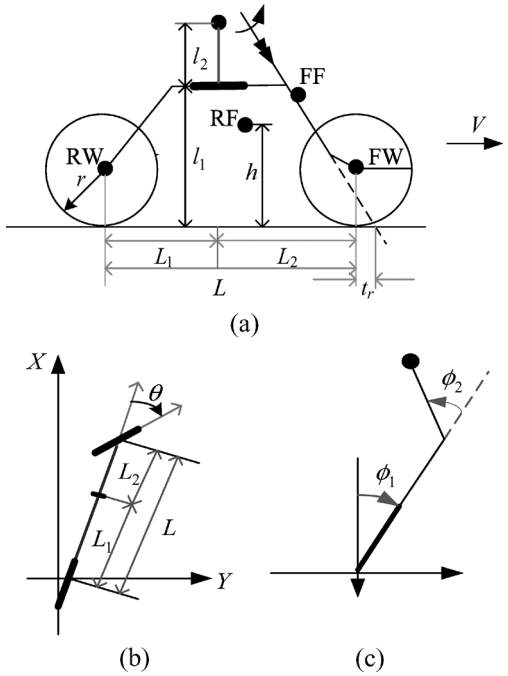


Fig. 1. Mechanism of an electrical bicycle. (a) Side view. (b) Top view. (c) Rear view.

$\dot{\phi}_2(t)$. Then, the corresponding dynamic equation in state-space form is achieved as follows:

$$
\begin{aligned}
\dot{x}_1(t) &= x_4(t) = a_1(x)\\
\dot{x}_2(t) &= x_5(t) = a_2(x)\\
\dot{x}_3(t) &= x_6(t) = a_3(x)\\
\dot{x}_4(t) &= \{f\ mg\sin x_2(t) + I_w\omega x_5(t)\\
&\quad - (C_t x_2(t) t_r \cos x_1(t) + \mu\omega t_r \sin x_1(t))\\
&\quad + \tau_{\rm han}(t)\}/I_h\\
&= a_4(x) + b_{41}(x)\tau_{\rm han}(t)\\
\dot{x}_5(t) &= \{-\left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad \times \left(M_p l_1 l_2 x_5^2 \sin x_{.3}(t)\right)\\
&\quad + \left(2M_p l_1 l_2 x_5(t) x_6(t) \sin x_3(t)\right)\left(M_p l_2^2 + J_2\right)\\
&\quad + \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad \times \left(M_p g l_2 \sin\left(x_3(t) - x_2(t)\right)\right)\\
&\quad + \left(M_p l_1 l_2 x_6^2(t) \sin x_3(t)\right)\left(M_p l_2^2 + J_2\right)\\
&\quad + \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad \cdot \left[M_p l_2 \cos\left(x_3(t) - x_2(t)\right)\right.\\
&\quad \left.\times \left(\frac{V^2}{L} x_1(t) + \frac{L_1 V}{L} x_4(t)\right)\right]\\
&\quad + \left(M_b g h \sin x_2(t)\right)\left(M_p l_2^2 + J_2\right)\\
&\quad + \left(M_p g l_1 \sin x_2(t)\right)\left(M_p l_2^2 + J_2\right)\\
&\quad - \left(M_p g l_2 \sin\left(x_3(t) - x_2(t)\right)\right)\left(M_p l_2^2 + J_2\right)\\
&\quad - [M_b l_1 \cos x_2(t) + M_p l_2 \cos x_2(t)\\
&\quad + M_p l_2 \cos\left(x_3(t) - x_2(t)\right)]\\
&\quad \cdot \left(\frac{V^2}{L} x_1(t) + \frac{L_1 V}{L} x_4(t)\right)\left(M_p l_2^2 + J_2\right)\\
&\quad + \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\tau_{\rm pen}(t)\}\\
&\quad /F(x_2, x_3)\\
&= a_5(x) + b_{52}(x)\tau_{\rm pen}(t)\\
\dot{x}_6(t) &= \{\left(M_b h^2 + M_p l_1^2 + M_p l_2^2\right.\\
&\quad \left. + 2M_p l_1 l_2 \cos x_3(t) + J_1 + J_2\right)\\
&\quad \cdot \left(M_p l_1 l_2 x_5^2(t) \sin x_3(t)\right)\\
&\quad - \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad \times \left(2M_p l_1 l_2 x_5(t) x_6(t) \sin x_3(t)\right)\\
&\quad - \left(M_b h^2 + M_p l_1^2 + M_p l_2^2\right.\\
&\quad \left. + 2M_p l_1 l_2 \cos x_3(t) + J_1 + J_2\right)\\
&\quad \times \left(M_p g l_2 \sin\left(x_3(t) - x_2(t)\right)\right)\\
&\quad - \left(M_p l_1 l_2 x_6^2 \sin x_3(t)\right)\\
&\quad \times \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad - \left(M_b h^2 + M_p l_1^2 + M_p l_2^2\right.\\
&\quad \left. + 2M_p l_1 l_2 \cos x_3(t) + J_1 + J_2\right)\\
&\quad \cdot \left[M_p l_2 \cos\left(x_3(t) - x_2(t)\right)\right.\\
&\quad \left.\times \left(\frac{V^2}{L} x_1(t) + \frac{L_1 V}{L} x_4(t)\right)\right]\\
&\quad - \left(M_b g h \sin x_2(t)\right)\\
&\quad \times \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad - \left(M_p g l_1 \sin x_2(t)\right)\\
&\quad \times \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad + \left(M_p g l_2 \sin\left(x_3(t) - x_2(t)\right)\right)\\
&\quad \times \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad + [M_b l_1 \cos x_2(t) + M_p l_2 \cos x_2(t)\\
&\quad + M_p l_2 \cos\left(x_3(t) - x_2(t)\right)]\\
&\quad \cdot \left(\frac{V^2}{L} x_1(t) + \frac{L_1 V}{L} x_4(t)\right)\\
&\quad \times \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)\\
&\quad - \left(M_b h^2 + M_p l_1^2 + M_p l_2^2\right.\\
&\quad \left. + 2M_p l_1 l_2 \cos x_3(t) + J_1 + J_2\right)\tau_{\rm pen}(t)\}\\
&\quad /G(x_3)\\
&= a_6(x) + b_{62}(x)\tau_{\rm pen}(t) \qquad \text{(1a)}
\end{aligned}
$$

where

$$
\begin{aligned}
F(x_3) &= \left(M_b h^2 + M_p l_1^2 + M_p l_2^2\right.\\
&\quad \left. + 2M_p l_1 l_2 \cos x_3(t) + J_1 + J_2\right)\cdot\left(M_p l_2^2 + J_2\right)\\
&\quad - \left(M_p l_2^2 + M_p l_1 l_2 \cos x_3(t) + J_2\right)^2 \qquad \text{(1b)}\\
G(x_3) &= -F(x_3) \qquad \text{(1c)}
\end{aligned}
$$

or the corresponding matrix form with uncertainty is written as follows:

$$\dot{x}(t) = a(x) + B(x)u(t) + c(x,t) \qquad (2)$$

where $x(t) = [x_1(t)\ \ x_2(t)\ \ x_3(t)\ \ x_4(t)\ \ x_5(t)\ \ x_6(t)]^{\rm T} \in \Re^6$ represents the system state which is available, $a(x) = [a_1(x)\ \ a_2(x)\ \ a_3(x)\ \ a_4(x)\ \ a_5(x)\ \ a_6(x)]^{\rm T}$ denotes a mapping from $\Re^6 \rightarrow \Re^6$, $B(x) =$

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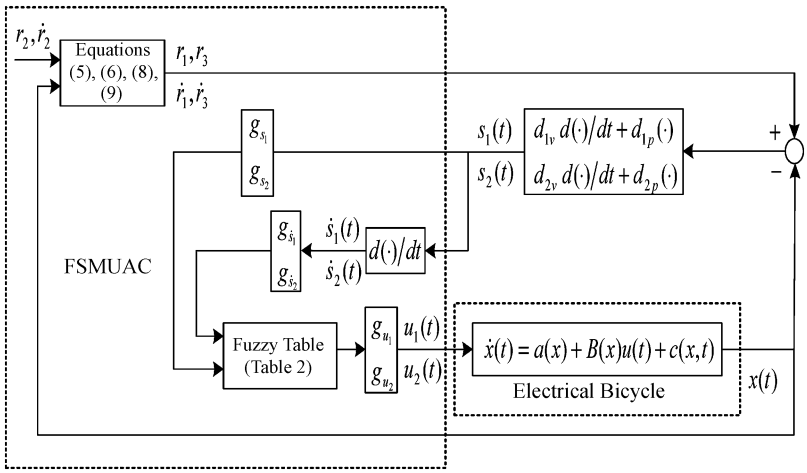


Fig. 2. Control block diagram of the overall system.

$\begin{bmatrix} 0 & 0 & 0 & b_{41}(x) & 0 & 0 \\ 0 & 0 & 0 & 0 & b_{52}(x) & b_{62}(x) \end{bmatrix}^{\mathrm{T}}$ denotes a mapping from $\Re^6 \to \Re^{6\times 2}$, $u(t) = [\tau_{\mathrm{han}}(t) \quad \tau_{\mathrm{pen}}(t)]^{\mathrm{T}} \in \Re^2$ represents the control torque, and $c(x,t)$ denotes nonlinear time-varying uncertainty caused by the parameter variations, e.g., $\Delta a_i(x,t)$, $i = 1,2,\ldots,6$, $\Delta b_{41}(x,t)$, $\Delta b_{52}(x,t)$, and $\Delta b_{62}(x,t)$ in a multiplicity form. After the examination of $b_{41}(x)$ and $b_{62}(x)$, they are all greater than zero for all $x(t)$. The corresponding simulation in Fig. 7(d) confirms the result.

B. Problem Formulation

Because the motion of an electrical bicycle is often in different operating conditions (e.g., concrete ground, muddy ground, wet ground, different tire pressures, and in the face of a gust of wind), the dynamics of an electrical bicycle is nonlinear and time variant. Hence, it is assumed that the controlled system, i.e., $a(x)$, $B(x)$, and $c(x,t)$, is unknown. However, the upper bounds of these functions are known. The control objective of this paper is to design the control torque $u(t)$, which is based on the upper bounds of $a(x)$, $B(x)$, and $c(x,t)$ such that the outputs $x_1(t) = \theta(t)$, $x_2(t) = \phi_1(t)$, and $x_3(t) = \phi_2(t)$ exponentially converge into a bounded set of zero and that the dynamic balance of the electrical bicycle is assured (cf. Fig. 2).

III. Properties of Underactuated Control

Because the proposed control is an underactuated control system, the following two sliding surfaces are first defined:

$$s_1(t) = d_{1v}\left(\dot{r}_1(t) - x_4(t)\right) + d_{1p}\left(r_1(t) - x_1(t)\right) \quad (3)$$

$$s_2(t) = d_{2v}\left(\dot{r}_3(t) - x_6(t)\right) + d_{2p}\left(r_3(t) - x_3(t)\right) \quad (4)$$

where $d_{1v}, d_{1p}, d_{2v}, d_{2p} > 0$ are assigned to obtain two stable sliding surfaces $s_1(t)$ and $s_2(t)$, and the reference inputs $r_1(t)$ and $r_3(t)$ are assigned as follows:

$$r_1(t) = k_{1v}\left(\dot{r}_2(t) - x_5(t)\right) + k_{1p}\left(r_2(t) - x_2(t)\right) \quad (5)$$

$$r_3(t) = k_{2v}\left(\dot{r}_2(t) - x_5(t)\right) + k_{2p}\left(r_2(t) - x_2(t)\right) \quad (6)$$

where $k_{1v}, k_{1p}, k_{2v}, k_{2p} > 0$ are set to obtain two stable reference inputs $r_1(t)$ and $r_3(t)$. The reason to design these reference signals of (5) and (6) is that the uncontrolled states $x_2(t)$ and $x_5(t)$ should be simultaneously included into the two control modes [i.e., $x_1(t), x_3(t)$ or $x_4(t), x_6(t)$] so that they are indirectly manipulated by these controllable states. For simplicity, the definitions of tracking error are described as follows:

$$e_i(t) = r_i(t) - x_i(t) \qquad \dot{e}_i(t) = e_{i+3}(t), \qquad i = 1,2,3. \quad (7)$$

Before discussing the properties of FSMUAC, the following two lemmas are given to depict the BIBO and asymptotical tracking.

Lemma 1 [25]: Consider the dynamic equation $a\dot{e}(t) + be(t) = s(t)$, where a and b are positive constants. If $|s(t)| < p$ for all $t \geq t_0$ and $e(t_0)$ is bounded, then $|e(t)| < p/b$ and $|\dot{e}(t)| < 2p/a$ as $t \to \infty$.

Lemma 2 [25]: Consider the dynamic equation $a\dot{e}(t) + be(t) = s(t)$, where a and b are positive constants. If $|s(t)| < b|e(t)|$ for all $t \geq t_0$ and $e(t_0)$ is bounded, then $\{e(t), \dot{e}(t)\}$ is bounded, and $|e(t)|, |\dot{e}(t)| \to 0$ as $t \to \infty$.

Proof: Using Bellman–Gronwall inequality lemma and triangular inequality obtains the result.

Then, the following theorems discuss the asymptotic and bounded tracking of a running electrical bicycle.

Theorem 1: If the sliding surfaces (3) and (4) have $|s_1(t)| < d_{1p}|e_1(t)|$ and $|s_2(t)| < d_{2p}|e_3(t)|$ as $t \geq t_0$ and the reference inputs satisfy the conditions $|r_1(t)|, |r_3(t)| < \max(k_{1p}, k_{2p})|e_2(t)|$ and $|\dot{r}_1(t)|, |\dot{r}_3(t)| < k_0|\dot{e}_2(t)|$, where k_0 is a bounded positive constant, as $t \geq t_0 q$, then $\{x_i(t)\}$ are bounded, and $|x_i(t)| \to 0$, $i = 1,2,\ldots,6$, as $t \to \infty$.

Proof: See Appendix B.

Theorem 2: Consider Theorem 1 except that the reference inputs satisfy the conditions $|r_1(t)| < p_1$, $|\dot{r}_1(t)| < p_2$, $|r_3(t)| < q_1$, and $|\dot{r}_3(t)| < q_2$ as $t \geq t_0$. Then, $|x_1(t)| < p_1$, $|x_2(t)| < \max(p_1/p_1k_{1p}, q_1/k_{2p})$, $|x_3(t)| < q_1$, $|x_4(t)| < p_2$, $|x_5(t)| < 2\max(p_1/k_{1v}, q_1/k_{2v})$, and $|x_6(t)| < q_2$ as $t \to \infty$.

Proof: See Appendix C.

Theorem 3: If the sliding surfaces (3) and (4) have $|s_1(t)| < p_0$ and $|s_2(t)| < q_0$ as $t \geq t_0$ and the reference inputs satisfy the condition $|r_1(t)| < p_1$, $|\dot{r}_1(t)| < p_2$, $|r_3(t)| < q_1$, and $|\dot{r}_3(t)| < q_2$ as $t \geq t_0$, then $|x_1(t)| < p_1 + p_0/d_{1p}$, $|x_2(t)| < \max(p_1/k_{1p}, q_1/k_{2p})$, $|x_3(t)| < q_1 + q_0/d_{2p}$, $|x_4(t)| < p_2 +$

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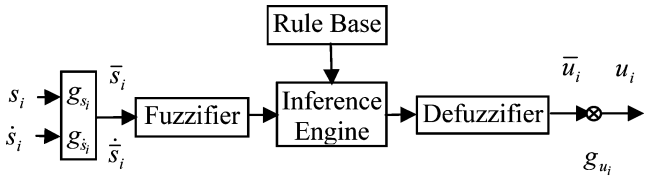


Fig. 3. Basic configuration of the ith fuzzy logic system.

p_0/d_{1v}, $|x_5(t)| < 2\max(p_1/k_{1v}, q_1/k_{2v})$, and $|x_6(t)| < q_2 + q_0/d_{2v}$ as $t \to \infty$.

Proof: See Appendix D.

The aforementioned lemmas and theorems are the fundamentals for the underactuated control.

IV. Controller Design

Based on (5) and (6) and (2)

$$\begin{aligned}\dot{r}_1(t) &= k_{1v}\left(\ddot{r}_2(t) - \dot{x}_5(t)\right) + k_{1p}\left(\dot{r}_2(t) - \dot{x}_2(t)\right)\\ &= k_{1v}\left[\ddot{r}_2(t) - a_5(x) - b_{52}(x)u_2(t) - c_5(x,t)\right]\\ &\quad + k_{1p}\left(\dot{r}_2(t) - x_5(t) - c_2(x,t)\right) \qquad (8)\\ \dot{r}_3(t) &= k_{2v}\left(\ddot{r}_2(t) - \dot{x}_5(t)\right) + k_{2p}\left(\dot{r}_2(t) - \dot{x}_2(t)\right)\\ &= k_{2v}\left[\ddot{r}_2(t) - a_5(x) - b_{52}(x)u_2(t) - c_5(x,t)\right]\\ &\quad + k_{2p}\left(\dot{r}_2(t) - x_5(t) - c_2(x,t)\right). \qquad (9)\end{aligned}$$

Then, the derivatives of these two sliding surfaces are expressed as follows:

$$\begin{aligned}\dot{s}_1(t) &= d_{1v}\left(\ddot{r}_1(t) - \dot{x}_4(t)\right) + d_{1p}\left(\dot{r}_1(t) - \dot{x}_1(t)\right)\\ &= d_{1v}\left[\ddot{r}_1(t) - a_4(x) - b_{41}(x)u_1(t) - c_4(x,t)\right]\\ &\quad + d_{1p}\left(\dot{r}_1(t) - x_4(t) - c_1(x,t)\right) \qquad (10)\\ \dot{s}_2(t) &= d_{2v}\left(\ddot{r}_3(t) - \dot{x}_6(t)\right) + d_{2p}\left(\dot{r}_3(t) - \dot{x}_3(t)\right)\\ &= d_{2v}\left[\ddot{r}_3(t) - a_6(x) - b_{62}(x)u_2(t) - c_6(x,t)\right]\\ &\quad + d_{2p}\left(\dot{r}_3(t) - x_6(t) - c_3(x,t)\right) \qquad (11)\end{aligned}$$

where $c_i(x,t)$, $i = 1,2,\ldots,6$, denotes the ith component of $c(x,t)$.

An important contribution of fuzzy systems theory is to provide a systematic procedure for transforming a set of linguistic rules into a nonlinear mapping. The fuzzy logic subsystem i for the FSMUAC in Fig. 3 performs a mapping from $X_i \in \Re^2$ to $\Re$. There are l fuzzy control rules, and the upper script k denotes the kth fuzzy rule

$$\text{IF } \bar{s}_i(t) \text{ is } F_{1_i}^k \text{ and } \dot{\bar{s}}_i(t) \text{ is } F_{2_i}^k, \quad \text{THEN } \bar{u}_i(t) \text{ is } G_i^k \qquad (12)$$

where $x_i(t) = [\,\bar{s}_i(t) \quad \dot{\bar{s}}_i(t)\,]^{\mathrm{T}} \in X_i \subset \Re^2$, with $\bar{s}_i(t) = g_{s_i}s_i(t)$ and $\dot{\bar{s}}_i(t) = g_{\dot{s}_i}\dot{s}_i(t)$, $\bar{u}_i(t) \in V_i \subset \Re$ are the input and output of the fuzzy logic subsystem i, respectively, and $F_{j_i}^k$ ($1 \le i,j \le 2, 1 \le k \le l$) and G_i^k are labels of sets in X_i and V_i, respectively. The parameters g_{s_i} and $g_{\dot{s}_i}$ are chosen such that $\bar{s}_i(t)$ and $\dot{\bar{s}}_i(t) \in [-1,1]$. The fuzzy inference engine performs a mapping from fuzzy sets in $X_i \subset \Re^2$ to fuzzy sets in $V_i \subset \Re$, based upon the fuzzy if–then rules in the fuzzy rule base and the compositional rule of inference. Let A_{x_i} be an arbitrary fuzzy set in X_i. The fuzzifier maps a crisp point $x_i(t)$ into a fuzzy set A_{x_i} in X_i. The center-average defuzzifier maps a fuzzy set in V_i to a crisp point in V_i.

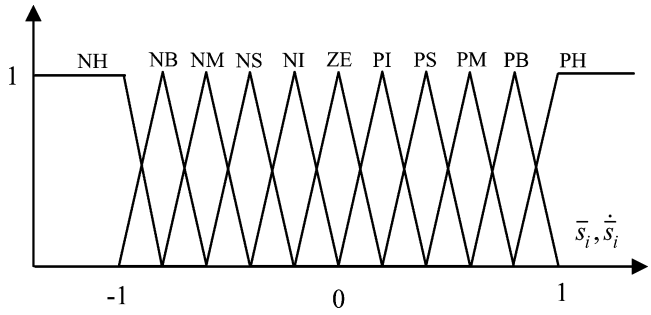


Fig. 4. Membership functions with triangular type.

TABLE III
Parameters of the Electrical Bicycle

Symbols	Description	Values
M_b	Mass of electrical bicycle	52.0 kg
m	Mass of front wheel part	10.0 kg
M_p	Mass of inverted pendulum	2.0 kg
L	Wheel base	1.13 m
L_1	CG position from rear wheel	0.324 m
L_2	CG position from front wheel	0.806 m
h	Height of CG	0.855 m
l_1	Height of the axis of pendulum	1.0 m
l_2	Length of pendulum	0.51 m
r	Radius of wheel	0.35 m
V	Velocity of bicycle	3.0 m/s
f	Offset	0.06 m
t_r	Trail	0.05 m
I_h	Inertial of front wheel about handle axis	0.35 kgm^2
I_w	Inertial of wheel	0.18 kgm^2
μ	Coefficient of conflict	0.1
J_1	Inertial of CG about x axis	10.0 kgm^2
J_2	Inertial of inverted pendulum	0.0463 kgm^2
C_t	Coefficient of camber thrust	66.0 N/rad

The output of the FSMUAC is then designed as follows:

$$u_i(t) = g_{u_i}\bar{u}_i(t) = g_{u_i}\left[s_i(t) + \Delta_i(t)\mathrm{sgn}(s_i)\right], \qquad i = 1,2 \qquad (13)$$

where $\bar{u}_i(t)$ is the fuzzy variable of $u(t)$ and $\Delta_i(t) > 0\ \forall t$ denotes the switching gain, which is related to Table I. It is also assumed that the output scaling factors satisfy the following inequalities:

$$\begin{aligned} g_{u_1} &\ge \left[|h_1(x,t)| + \lambda_1\right] / \left[d_{1v}b_{41}(x)\Delta_1(t)\right]\\ g_{u_2} &\ge \left[|h_2(x,t)| + \lambda_2\right] / \left[d_{2v}b_{62}(x)\Delta_2(t)\right] \qquad (14)\end{aligned}$$

where

$$\begin{aligned} h_1(x,t) &= d_{1v}\left(\ddot{r}_1(t) - a_4(x) - c_4(x,t)\right)\\ &\quad + d_{1p}\left(\dot{r}_1(t) - x_4(t) - c_1(x,t)\right) \qquad (15)\\ h_2(x,t) &= d_{2v}\left(\ddot{r}_3(t) - a_6(x) - c_6(x,t)\right)\\ &\quad + d_{2p}\left(\dot{r}_3(t) - x_6(t) - c_3(x,t)\right). \qquad (16)\end{aligned}$$

Before discussing the result of the controller, the following property about "uniformly ultimately bounded (UUB)" is also given.

Definition 1 [26]: The solutions of a dynamic system are said to be UUB if there exist positive constants υ and κ and, for every

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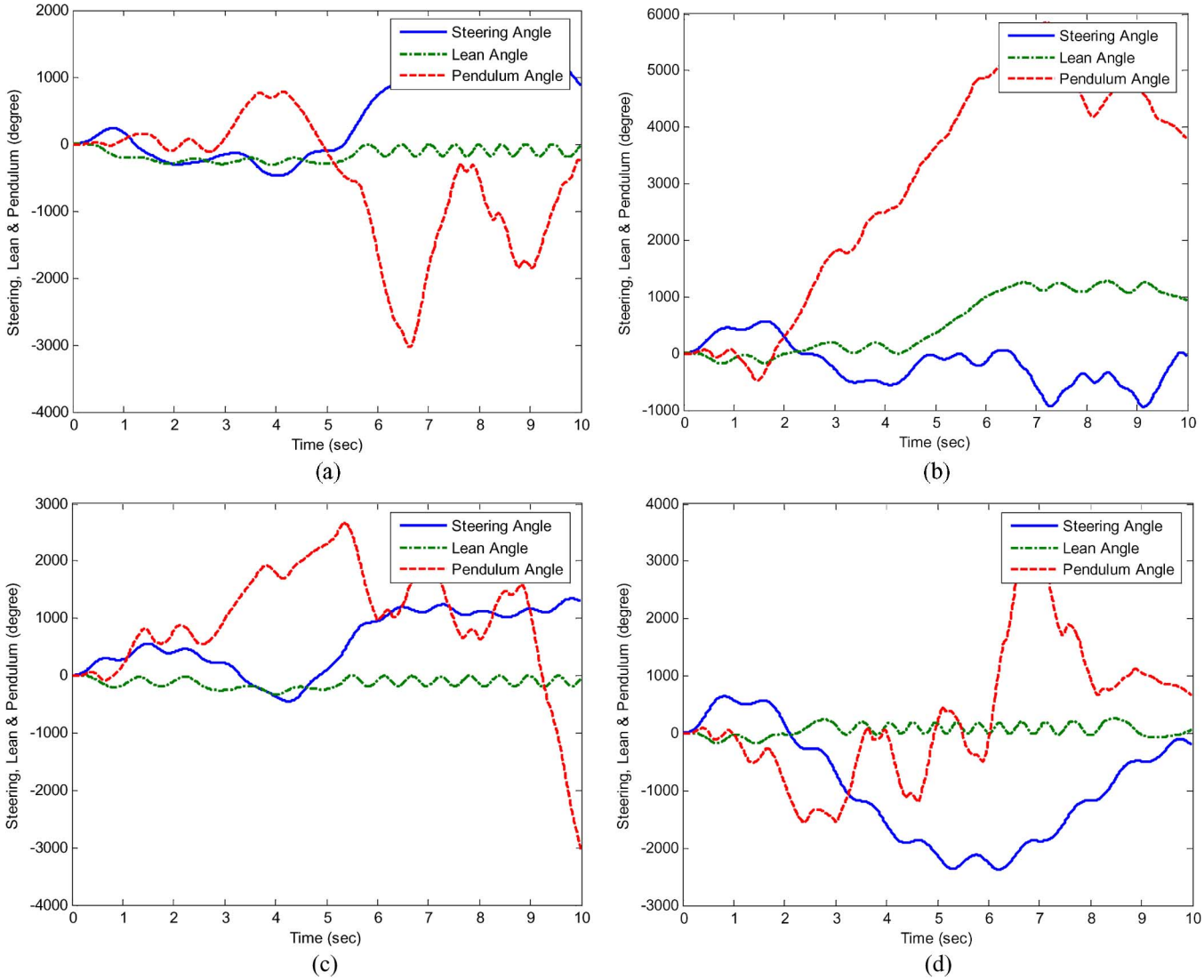


Fig. 5. Sinusoidal responses of the electrical bicycle. (a) $\tau_{\rm han}(t) = 10\sin(1.6\pi t)$ and $\tau_{\rm pen}(t) = 10\sin(1.6\pi t)$ N · m. (b) $\tau_{\rm han}(t) = 20\sin(2.0\pi t)$ and $\tau_{\rm pen}(t) = 20\sin(2.0\pi t)$ N · m. (c) $\tau_{\rm han}(t) = 20\sin(2.4\pi t)$ and $\tau_{\rm pen}(t) = 20\sin(2.4\pi t)$ N · m. (d) $\tau_{\rm han}(t) = 30\sin(2.0\pi t)$ and $\tau_{\rm pen}(t) = 30\sin(2.0\pi t)$ N · m.

$\delta \in (0, \kappa)$, there is a positive constant $T = T(\delta)$, such that $\|x(t_0)\| < \delta \Rightarrow \|x(t)\| \leq \upsilon \; \forall t \geq t_0 + T$.

The following theorem discusses the property of the electrical bicycle with the FSMUAC (13).

Theorem 4: Consider the unknown electrical bicycle (2) with the known upper bound of (14). The reference inputs satisfy the conditions $|r_1(t)|, |r_3(t)| < \max(k_{1p}, k_{2p})|e_2(t)|$ and $|\dot{r}_1(t)|, |\dot{r}_3(t)| < k_0|\dot{e}_2(t)|$, where k_0 is a bounded positive constant, as $t \geq t_0$. Applying (13) to system (2) gives the finite time to reach the sliding surfaces (3) and (4); $x_i(t)$ and $u_j(t)$, for $i = 1, 2, \ldots, 6$ and $j = 1, 2$, are UUB, and $|x_i(t)| \to 0$, $i = 1, 2, \ldots, 6$, as $t \to \infty$.

Proof: See Appendix E.

Theorem 5: Assume that the inequality (14) is satisfied outside of the following convex set:

$$D = \{s_1(t), s_2(t) \,|\, |s_1(t)| < p_0, |s_2(t)| < q_0\}.$$

In addition, the reference inputs satisfy the condition $|r_1(t)| < p_1$, $|\dot{r}_1(t)| < p_2$, $|r_3(t)| < q_1$, and $|\dot{r}_3(t)| < q_2$ as $t \geq t_0$. Then, $x_i(t)$ and $u_j(t)$, for $i = 1, 2, \ldots, 6$ and $j = 1, 2$, are UUB, and $|x_1(t)| < p_1 + p_0/d_{1p}$, $|x_2(t)| < \max(p_1/k_{1p}, q_1/k_{2p})$, $|x_3(t)| < q_1 + q_0/d_{2p}$, $|x_4(t)| < p_2 + p_0/d_{1v}$, $|x_5(t)| < 2\max(p_1/k_{1v}, q_1/k_{2v})$, and $|x_6(t)| < q_2 + q_0/d_{2v}$ as $t \to \infty$.

Proof: The corresponding result can be obtained from Theorems 3 and 4.

The corresponding three scaling factors (or control parameters) g_{s_i}, $g_{\dot{s}_i}$, and g_{u_i} are discussed as follows. Inequality (14) implies that the output scaling factor should be greater than the upper bound of system gains, control gains, and uncertainty. Due to the unstable feature of an electrical bicycle, a smallest g_{u_i} must be used to stabilize its dynamics. In the beginning, it is chosen from a suitable small value. According to the response, a larger value is then applied to improve the system performance. A larger output scaling factor is chosen, and a smaller tracking error is achieved. However, the risk of transient (or unstable) response occurs due to the constraints of the (rate of) control input. In addition, the oscillatory response after transience often happens.

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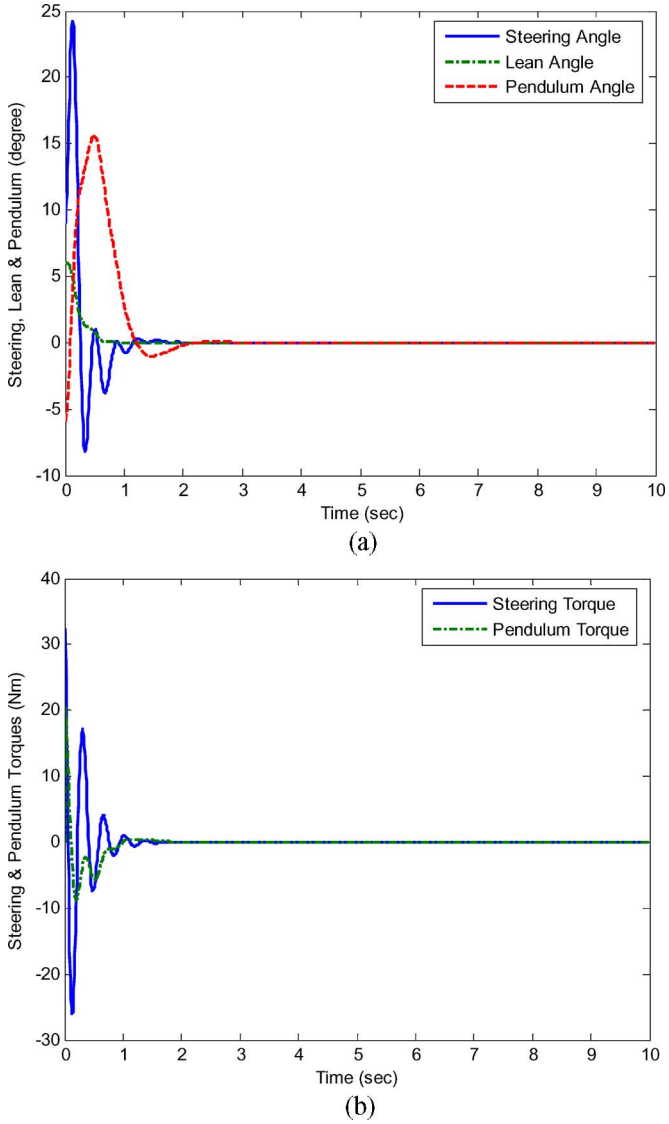


Fig. 6. Responses of the electrical bicycle in the absence of uncertainty by the MPDC with $d_{1v} = 1.8$, $d_{1p} = 18$, $d_{2v} = 3.0$, and $d_{2p} = 18$. (a) System outputs $\theta(t)$, $\phi_1(t)$, and $\phi_2(t)$. (b) Control inputs $\tau_{\text{han}}(t)$ and $\tau_{\text{pen}}(t)$.

Based on the input–output data, it is assumed that $\dot{s}_i(t)$ increases as $u_i(t) = g_{u_i}\bar{u}_i(t)$ decreases, and if $s_i(t) > 0$, then increasing $u_i(t)$ will result in decreasing $s_i(t)\dot{s}_i(t)$, and if $s_i(t) < 0$, then decreasing $u_i(t)$ will result in decreasing $s_i(t)\dot{s}_i(t)$. That is, the control input $u_i(t)$ is designed in an attempt to satisfy the inequality $s_i(t)\dot{s}_i(t) < 0, i = 1, 2$, which results in the decrease of Lyapunov function $V(t) = \sum_{i=1}^{2} s_i^2(t)/2$, i.e., $\dot{V}(t) < 0$.

Based on the aforementioned analysis, the fuzzy variables $\bar{s}_i(t) = g_{s_i}s_i(t)$ and $\dot{\bar{s}}_i(t) = g_{\dot{s}_i}\dot{s}_i(t)$, $i = 1, 2$, are quantized into the following 11 qualitative fuzzy variables (i.e., $l = 11$): 1) positive huge (PH); 2) positive big (PB); 3) positive medium (PM); 4) positive small (PS); 5) positive infinitesimal (PI); 6) zero (ZE); 7) negative infinitesimal (NI); 8) negative small (NS); 9) negative medium (NM); 10) negative big (NB); and 11) negative huge (NH). It is not necessary that 11 fuzzy rules must be selected. A smaller fuzzy rule (e.g., $l = 7$) can also obtain an acceptable performance (e.g., [7]). There are many types of membership functions, some of which are bell shaped, trapezoidal shaped, triangular shaped, etc. The triangular type in Fig. 4 is used in this paper. In summary, the linguistic rule of the ith FSMUAC is shown in Table I by which the CG method is employed to form a lookup table (Table II) that directly relates the inputs $\bar{s}_i(t)$ and $\dot{\bar{s}}_i(t)$ with the output $\bar{u}_i(t)$. The control actions of the diagonal terms in Table II are ZE. This arrangement is similar to a sliding-mode controller that has a sliding surface. In addition, the control actions of the upper triangle terms are from NI to NH, and those of the lower triangle terms are from PI to PH. In brief, it is skew symmetric.

V. Simulations and Discussions

Consider the electrical bicycle (2), as shown in Table III. The corresponding sinusoidal responses for some representative amplitudes and frequencies are shown in Fig. 5, which indicates that the dynamics of an electrical bicycle is highly nonlinear and unstable [18]–[24]. The ordinary PID control cannot stabilize the electrical bicycle. For brevity, these responses are omitted. However, it can be stabilized by the following modified proportional–derivative control (MPDC) (e.g., [21]):

$$u_{1\text{mpd}}(t) = d_{1v}\left(\dot{r}_1(t) - x_4(t)\right) + d_{1p}\left(r_1(t) - x_1(t)\right) \quad (17a)$$

$$u_{2\text{mpd}}(t) = d_{2v}\left(\dot{r}_3(t) - x_6(t)\right) + d_{2p}\left(r_3(t) - x_3(t)\right) \quad (17b)$$

where $r_1(t)$, $\dot{r}_1(t)$, $r_3(t)$, and $\dot{r}_3(t)$ are obtained from (5), (6), (8), and (9), respectively, and the parameters $k_{1v} = 3.0$, $k_{1p} = 20$, $k_{2v} = 0.1$, and $k_{2p} = 10$. In this paper, the reference inputs of the lean angle and velocity and the reference inputs of the nominal steering and pendulum acceleration are assigned to be zero, i.e., $r_2(t) = 0$, $\dot{r}_2(t) = 0$, $\ddot{r}_1(t) = 0$, and $\ddot{r}_3(t) = 0$. For example, the acceptable response of the electrical bicycle in the absence of uncertainty by the MPDC (17) with $d_{1v} = 1.8$, $d_{1p} = 18$, $d_{2v} = 3.0$, and $d_{2p} = 18$ is shown in Fig. 6. The responses of other suitable parameters are similar with those in Fig. 6. For simplicity, those are left out. For comparison between the MPDC and the proposed FSMUAC, the coefficients of the sliding surface are the same as the parameters of the MPDC. The uncertainty of the multiplication form (i.e., $a_i(x)(1 + \Delta a_i(x,t))$, $i = 4, 5, 6$, $b_{41}(x)(1 + \Delta b_{41}(x,t))$, $b_{52}(x)(1 + \Delta b_{52}(x,t))$, and $b_{62}(x)(1 + \Delta b_{62}(x,t))$) to represent the effect of $c(x,t)$ is assumed to be

$$\begin{aligned}
\Delta a_4(x,t) &= 1.5x_1(t)x_2(t)\sin\left(0.3tx_5(t)\right) \\
&\quad - 0.37x_3(t)x_6(t)\cos\left(0.2tx_6(t)\right) \\
&\quad - 0.2\sin(100t) \\
\Delta a_5(x,t) &= 1.5x_2(t)x_3(t)\sin\left(0.3tx_6(t)\right) \\
&\quad - 0.37x_1(t)x_4(t)\cos\left(0.1tx_5(t)\right) \\
&\quad - 0.2\cos(200t) \\
\Delta a_6(x,t) &= 1.5x_3(t)x_1(t)\sin\left(0.3tx_4(t)\right) \\
&\quad + 0.37x_2(t)x_5(t)\cos\left(0.3tx_4(t)\right) \\
&\quad + 0.2\sin(300t) \\
\Delta b_{41}(x,t) &= 0.3\sin\left(-0.1 + 9.6tx_3(t)\right) \\
&\quad + 0.5\sin(300t) \\
\Delta b_{52}(x,t) &= 0.3\sin\left(-0.1 + 10.7tx_2(t)\right) \\
&\quad + 0.5\sin(200t) \\
\Delta b_{62}(x,t) &= 0.3\sin\left(0.1 + 6.9tx_1(t)\right) \\
&\quad - 0.5\cos(100t). \qquad (18)
\end{aligned}$$

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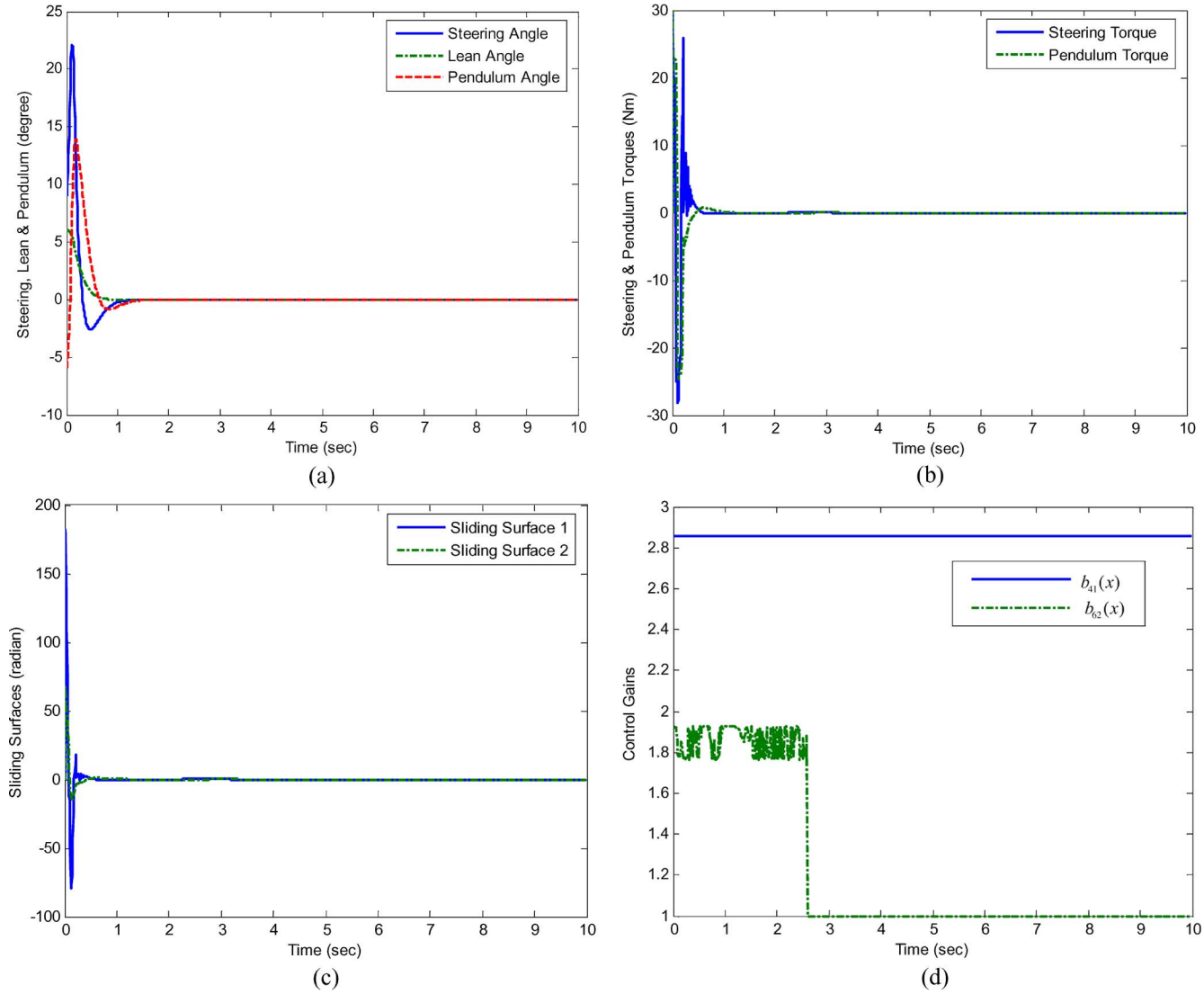


Fig. 7. Responses of the electrical bicycle in the absence of uncertainty by the FSMUAC. (a) System outputs $\theta(t)$, $\phi_1(t)$, and $\phi_2(t)$. (b) Control inputs $\tau_{\rm han}(t)$ and $\tau_{\rm pen}(t)$. (c) Sliding surfaces $s_1(t)$ and $s_2(t)$. (d) Control gains $b_{41}(x)$ and $b_{62}(x)$.

This uncertainty is nonlinear time varying and large. The following simulations use the numerical algorithm of the fourth-order Runge–Kutta with a time step of 0.01 s. The control parameters of the FSMUAC are set as follows: $g_{s_1} = 0.1$, $g_{\dot{s}_1} = 0.01$, $g_{u_1} = 30$, $g_{s_2} = 0.1$, $g_{\dot{s}_2} = 0.01$, and $g_{u_2} = 30$. The response for the electrical bicycle in the absence of uncertainty by the FSMUAC is shown in Fig. 7. Fig. 7(d) confirms that $b_{41}(x)$ and $b_{62}(x) > 0\ \forall x(t)$. It indicates that the performance of the MPDC (i.e., Fig. 6) is indeed improved by the proposed FSMUAC. Because the control system of an electrical bicycle is often encountered with large uncertainties, e.g., caused by different ground conditions (i.e., wet ground, muddy ground, concrete ground, and different tire pressures) and a gust of wind outside of the house, these responses must be examined before executing an experiment. Therefore, the responses of an electrical bicycle in the face of the uncertainty (18) by the FSMUAC are shown in Fig. 8. Simultaneously, the responses of an electrical bicycle in the presence of the uncertainty (18) by the MPDC are shown in Fig. 9. These two figures show that the robustness of the proposed FSMUAC is better than that of the MPDC.

For the consideration of practical situations (e.g., a gust of wind is blown to the bicycle), the following simulation with the uncertainty (18) is investigated. It is assumed that a gust of wind during the period of 4.0–4.1 s brings about the extra effect of the lean angle of 5 ° from 4.0 to 4.1 s. The corresponding result by the FSMUAC is then shown in Fig. 10. For comparison with the result of the MPDC, its relative response is also shown in Fig. 11. From these two figures, we know that the resistance of the pulse disturbance by the FSMUAC is superior to that of the MPDC. From a practical viewpoint, the following simulation with the saturations $|\tau_{\rm han}(t)|$ and $|\tau_{\rm pen}(t)| < \tau_{\max}$, where $\tau_{\max} = 20$ and 25 N · m, are also investigated. The responses for without saturation and saturations 20 and 25 are very similar. Hence, these responses are omitted. Finally, the responses in Fig. 10 with the saturated control input of 25 N·m is shown in Fig. 12. The related response is also acceptable. As one knows, a bicycle can be manipulated in a variable speed. Hence, the cor-

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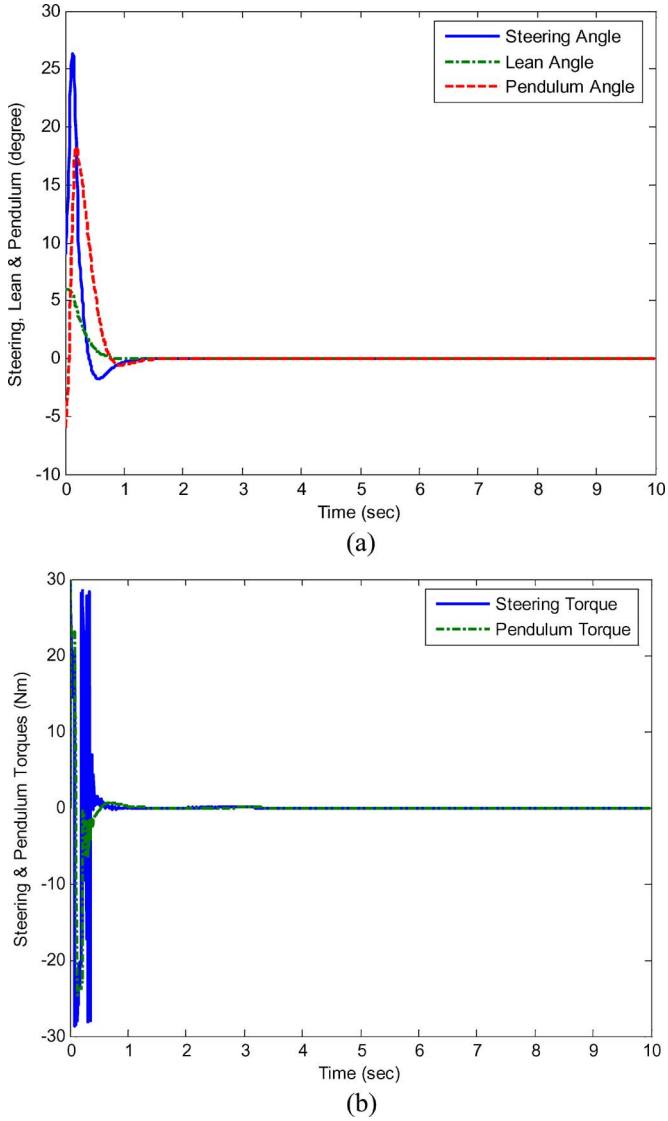


Fig. 8. Responses of the electrical bicycle in the face of the uncertainty (18) by the FSMUAC. (a) System outputs $\theta(t)$, $\phi_1(t)$, and $\phi_2(t)$. (b) Control inputs $\tau_{\rm han}(t)$ and $\tau_{\rm pen}(t)$.

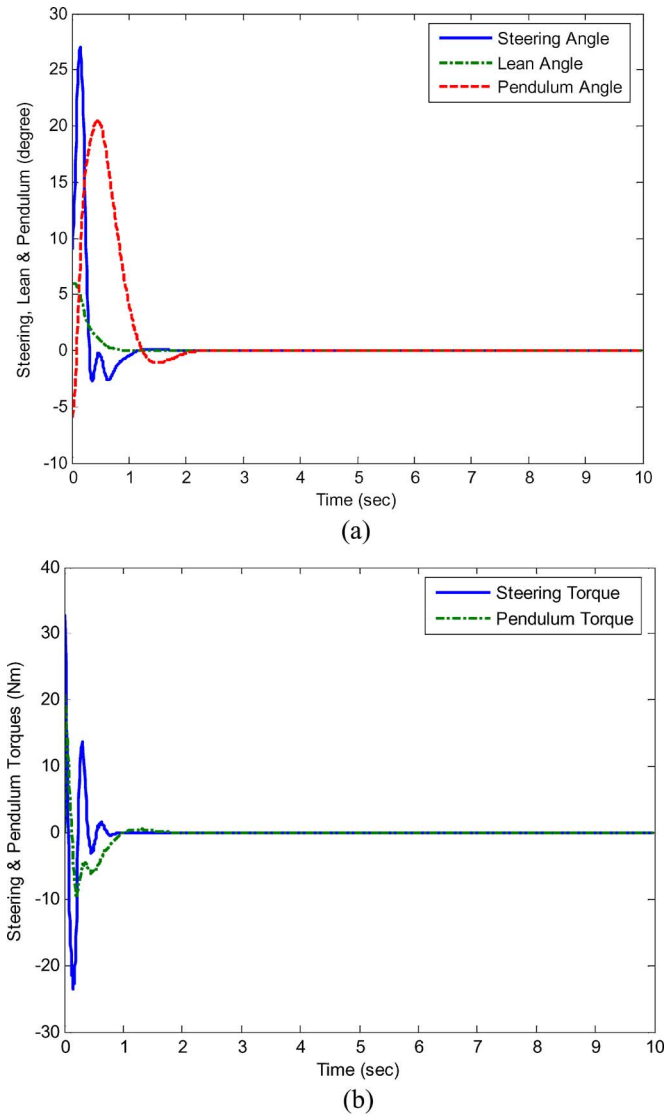


Fig. 9. Responses of the electrical bicycle in the face of the uncertainty (18) by the MPDC. (a) System outputs $\theta(t)$, $\phi_1(t)$, and $\phi_2(t)$. (b) Control inputs $\tau_{\rm han}(t)$ and $\tau_{\rm pen}(t)$.

responding result is examined by Figs. 13 and 14. Fig. 13 shows that the FSMUAC and MPDC can deal with a variable speed with 0.1 Hz from 0 to 6 m/s. To further understand the effect of variable speed in the motion of an electrical bicycle, the case of Fig. 13 with a pulse disturbance from 4.0 to 4.1 s and a larger variation of speed [e.g., $V(t) = 4 + 4\sin(0.2\pi t)$] is shown in Fig. 14. In Fig. 14, the speed is zero at a time of 7.5 s and then increases from 0 to 4 m/s for the time period from 7.5 to 10 s. Due to the existence of unmodeled dynamics (e.g., the uncertainty (18) and the pulse disturbance for the lean angle of 5 ° from 4.0 to 4.1 s), the MPDC becomes poor as the time-varying speed is in the neighborhood of zero [see Fig. 14(b)].

Finally, an implementation of the electrical bicycle is applied by an embedded control system, e.g., TMS320F2812 from TI Co. Can the control cycle time be larger than the 10 ms of the aforementioned setting? Based on this requirement, the corresponding results of the case in Fig. 10 using the control cycle times of 100 and 50 ms are addressed. The response using the control cycle time of 100 ms is unstable for the original setting. The response using the control cycle time of 50 ms becomes sluggish. For simplicity, these are omitted.

Based on the aforementioned investigations, the important observations are depicted as follows.

1) The dynamics of the electrical bicycle is nonlinear, complex, time variant, and unstable.
2) The FSMUAC can improve the performances of the MPDC and ordinary PID control.
3) A saturated control input of the FSMUAC does not necessarily result in a poor performance. However, too large saturation for the control input often brings about an oscillatory response.
4) The control cycle time for the electrical bicycle in motion should not be too large. In our case, the control cycle time larger than 100 ms will result in an inability.
5) The resistance of the pulse disturbance (e.g., a gust of wind) by the FSMUAC is better than that of the MPDC.

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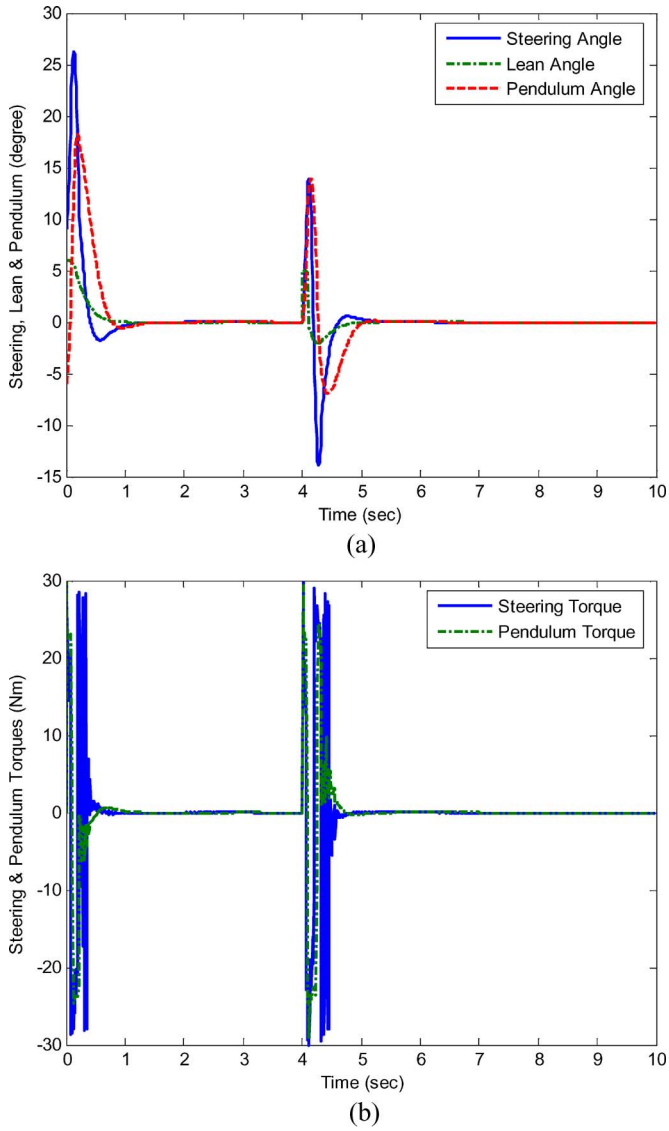


Fig. 10. Responses of Fig. 8 in the presence of the pulse disturbance of a lean angle of 5 ° from 4.0 to 4.1 s. (a) System outputs $\theta(t)$, $\phi_1(t)$, and $\phi_2(t)$. (b) Control inputs $\tau_{\rm han}(t)$ and $\tau_{\rm pen}(t)$.

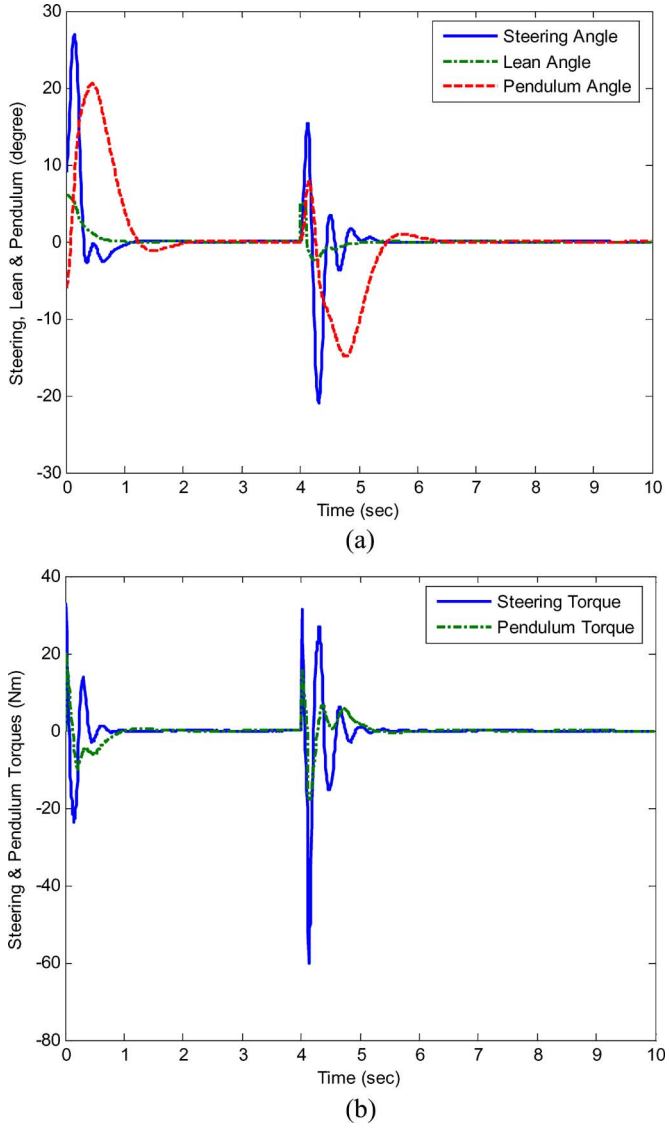


Fig. 11. Responses of Fig. 9 in the face of the pulse disturbance at a lean angle of 5 ° from 4.0 to 4.1 s. (a) System outputs $\theta(t)$, $\phi_1(t)$, and $\phi_2(t)$. (b) Control inputs $\tau_{\rm han}(t)$ and $\tau_{\rm pen}(t)$.

6) The proposed FSMUAC can be applied to the variable speed of the electrical bicycle (e.g., from 0 to 6 (or 8) m/sec with 0.1 Hz). On the other hand, the MPDC is not appropriate for the control of an electrical bicycle with this time-varying speed due to the existence of unmodeled dynamics [see Fig. 14(b)].

VI. Conclusion

The autonomous dynamic balance control of an electrical bicycle in motion provides feedbacks on the steering angle of the bicycle's handle, the lean angle of the bicycle, and the angle of a pendulum. On the other hand, the proposed FSMUAC merely provides steering and pendulum control torques. It is the first time to investigate the autonomous dynamic balance of a running electrical bicycle using FSMUAC. The important results are summarized as follows: 1) The properties of underactuated control is developed. Under suitable conditions, the bounded and asymptotical tracking results are obtained. 2) Only the data of system input/output and the information of the upper bound of system knowledge are required for the selection of three appropriate scaling factors. These include two normalizing scaling factors for the sliding surface and its derivative and one output scaling factor for the crisp control input. 3) The output scaling factor is determined by the system stability. A larger output scaling factor is assigned, and a smaller tracking error and a faster response are achieved; however, the risk of transient (or unstable) response happens. 4) The system performances, including the tracking performance for nonlinear time-varying uncertainty, generation of pulse disturbance to simulate wind effects, the saturation of control input due to the limitation of signals, and the control of the bicycle's varying speed motion, are improved as compared with ordinary PID control and the MPDC. 5) The stability of the closed-loop system is also verified by the Lyapunov stability theory.

Because of the advantageous features of the proposed FSMUAC, our future work is to implement this study.

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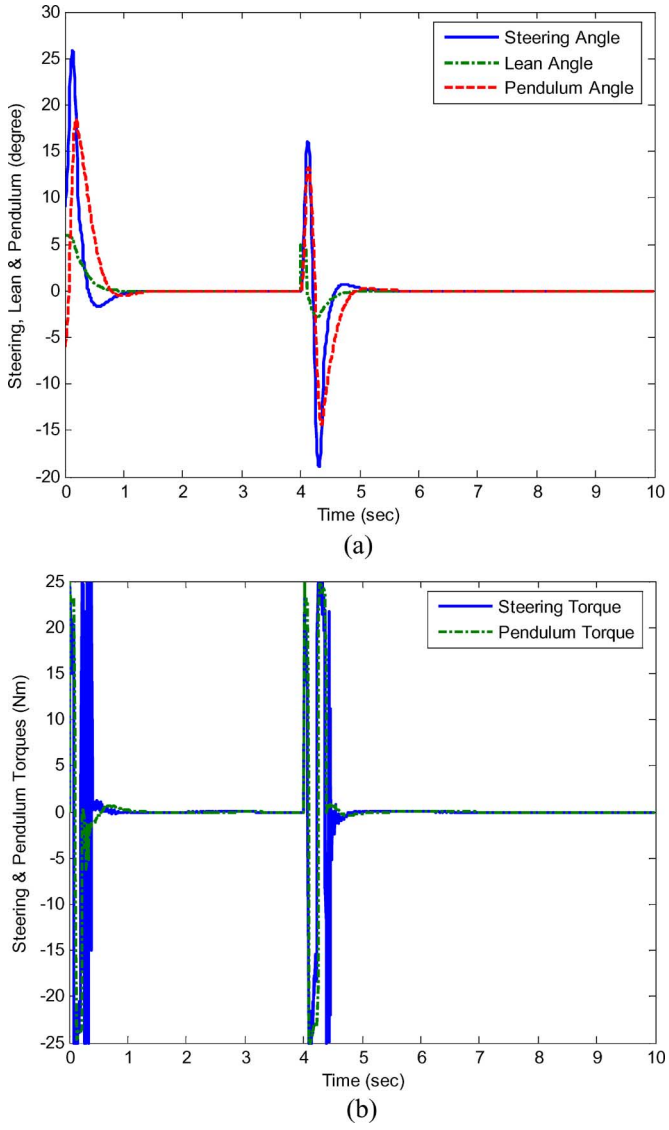


Fig. 12. Responses of the case in Fig. 10 with saturated control input. (a) System outputs $\theta(t)$, $\phi_1(t)$, and $\phi_2(t)$. (b) Control inputs $\tau_{\rm han}(t)$ and $\tau_{\rm pen}(t)$.

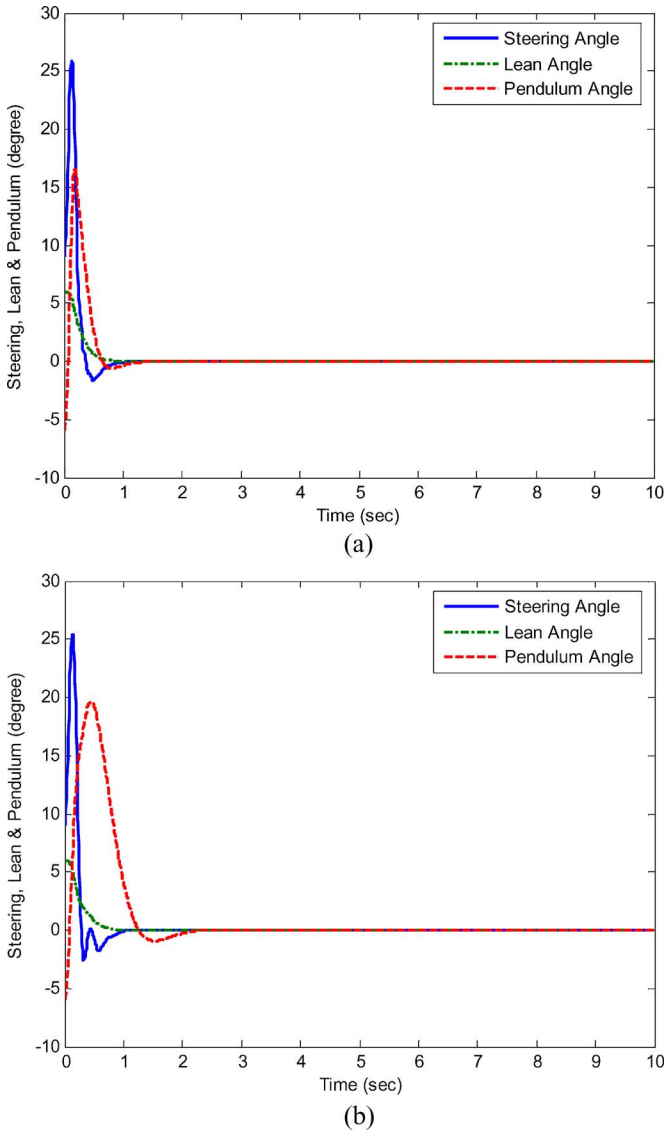


Fig. 13. Output responses of the electrical bicycle with variable speed $V(t) = 3 + 3\sin(0.2\pi t)$ in the face of the uncertainty (18) by (a) FSMUAC and (b) MPDC.

Appendix A
Dynamics of an Electrical Bicycle

The dynamic equation of the front wheel parties is first described as follows:

$$I_h\ddot{\theta}(t) = \tau_{\rm han}(t) + fmg\sin\phi_1(t) + I_w\omega\dot{\phi}_1(t) - (C_t\phi_1(t)t_r\cos\theta(t) + \mu\omega t_r\sin\theta(t)) \quad \text{(A1)}$$

where $\theta(t)$ and $\phi_1(t)$ are denoted as the steering angle and the lean angle, respectively, $\tau_{\rm han}(t)$ denotes the torque applied to the handle, and the other parameters are explained and shown in Table III. Then, the dynamic equation of the bicycle body part is given as follows:

$$\begin{aligned}&\left(M_bh^2 + M_pl_1^2 + M_pl_2^2 + 2M_pl_1l_2\cos\phi_2(t) + J_1 + J_2\right)\\ &\quad\times\ddot{\phi}_1(t) - \left(M_pl_2^2 + M_pl_1l_2\cos\phi_2(t) + J_2\right)\ddot{\phi}_2(t)\\ &\quad - 2M_pl_1l_2\dot{\phi}_1(t)\dot{\phi}_2(t)\sin\phi_2(t) + M_pl_1l_2\dot{\phi}_2^2(t)\\ &\quad\times\sin\phi_2(t) - M_bgh\sin\phi_1(t) - M_pgl_1\sin\phi_1(t)\\ &\quad + M_pgl_2\sin\left(\phi_2(t) - \phi_1(t)\right)\\ &\quad + \left(M_bl_1\cos\phi_1(t) + M_pl_2\cos\phi_1(t)\right.\\ &\quad \left. + M_pl_2\cos\left(\phi_2(t) - \phi_1(t)\right)\right)\left(\frac{V^2}{L}\theta(t) + \frac{L_1V}{L}\dot{\theta}(t)\right) = 0\end{aligned} \quad \text{(A2)}$$

$$\begin{aligned}&-\left(M_pl_2^2 + M_pl_1l_2\cos\phi_2(t) + J_2\right)\ddot{\phi}_1(t)\\ &\quad + \left(M_pl_2^2 + J_2\right)\ddot{\phi}_2(t) + M_pl_1l_2\dot{\phi}_1^2(t)\sin\phi_2(t)\\ &\quad - M_pgl_2\sin\left(\phi_2(t) - \phi_1(t)\right)\\ &\quad - M_pl_2\cos\left(\phi_2(t) - \phi_1(t)\right)\left(\frac{V^2}{L}\theta(t) + \frac{L_1V}{L}\dot{\theta}(t)\right) = \tau_{\rm pen}(t)\end{aligned} \quad \text{(A3)}$$

where $\phi_2(t)$ is the pendulum angle, $\tau_{\rm pen}(t)$ is the torque applied to the pendulum, and the other parameters are also explained and shown in Table III. Based on (A1), (A2), and (A3), the corresponding state equation (1) is obtained. Q.E.D.

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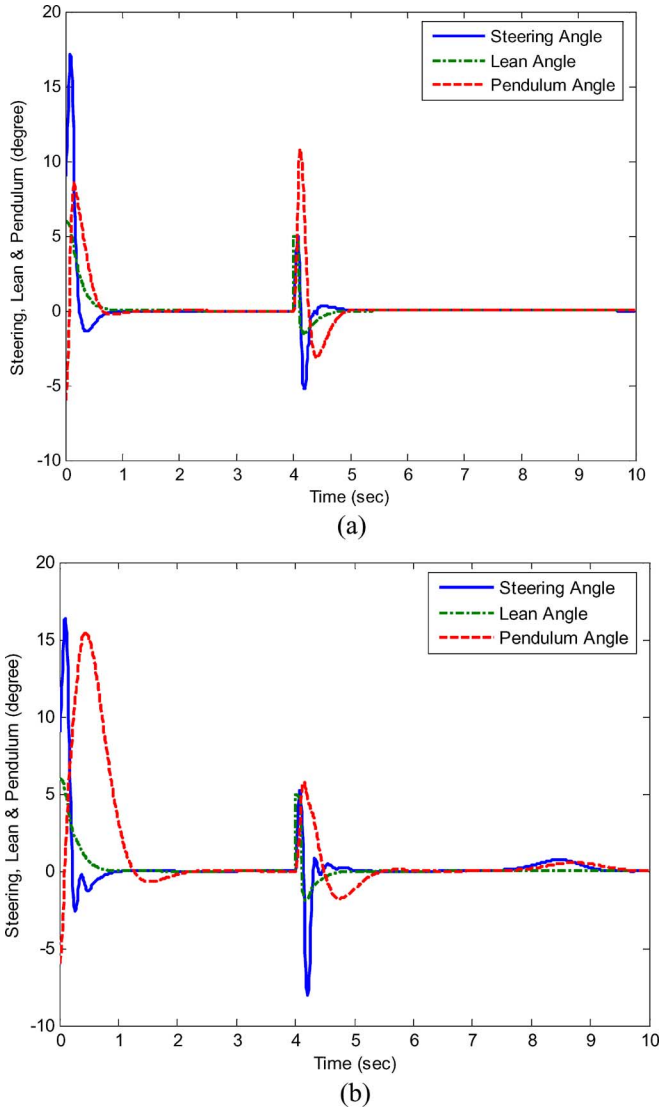


Fig. 14. Output responses of the electrical bicycle with variable speed $V(t) = 4 + 4\sin(0.2\pi t)$ in the face of the uncertainty (18) and the pulse disturbance from 4.0 to 4.1 s by (a) FSMUAC and (b) MPDC.

Appendix B
Proof of Theorem 1

Applying Lemma 2 to (3) and (4) with the fact that $|s_1(t)| < d_{1p}|e_1(t)|$ and $|s_2(t)| < d_{2p}|e_3(t)|$, then $\{e_1(t), \dot{e}_1(t), e_3(t), \dot{e}_3(t)\}$ are bounded, and $|e_1(t)|, |\dot{e}_1(t)|, |e_3(t)|, |\dot{e}_3(t)| \to 0$ as $t \to \infty$. Similarly, based on the results $|r_1(t)|, |r_3(t)| < \max(k_{1p}, k_{2p})|e_2(t)|$ and Lemma 2, $\{e_2(t), \dot{e}_2(t)\}$ is bounded, and $|e_2(t)|, |\dot{e}_2(t)| \to 0$ as $t \to \infty$. Because $r_2(t) = \dot{r}_2(t) = 0$, $|x_2(t)|, |x_5(t)| \to 0$ as $t \to \infty$. Then, from the inequalities $|r_1(t)|, |r_3(t)| < \max(k_{1p}, k_{2p})|e_2(t)|$ and $|\dot{r}_1(t)|, |\dot{r}_3(t)| < k_0|\dot{e}_2(t)|$, $r_1(t), \dot{r}_1(t), r_3(t), \dot{r}_3(t) \to 0$ as $t \to \infty$. Hence, the result that $|x_1(t)|, |x_3(t)|, |x_4(t)|, |x_6(t)| \to 0$ as $t \to \infty$ is derived from $|e_1(t)|, |\dot{e}_1(t)|, |e_3(t)|, |\dot{e}_3(t)| \to 0$ as $t \to \infty$. Q.E.D.

Appendix C
Proof of Theorem 2

Based on Lemma 2, $\{e_1(t), \dot{e}_1(t), e_3(t), \dot{e}_3(t)\}$ are bounded, and $|e_1(t)|, |\dot{e}_1(t)|, |e_3(t)|, |\dot{e}_3(t)| \to 0$ as $t \to \infty$. Applying Lemma 1 to (5) and (6) with the satisfaction of $|r_1(t)| < p_1$ and $|r_3(t)| < q_1$ obtains $|e_2(t)| < p_1/k_{1p}$, $|\dot{e}_2(t)| < 2p_1/k_{1v}$ and $|e_2(t)| < q_1/k_{2p}$, $|\dot{e}_2(t)| < 2q_1/k_{2v}$, respectively. Finally, based on the definitions of (7) and the triangular inequality, the corresponding results are obtained. Q.E.D.

Appendix D
Proof of Theorem 3

Applying Lemma 1 to (3) and (4) with $|s_1(t)| < p_0$ and $|s_2(t)| < q_0$, respectively, gives $|e_1(t)| < p_0/d_{1p}$, $|\dot{e}_1(t)| < 2p_0/d_{1v}$ and $|e_3(t)| < q_0/d_{1p}$, $|\dot{e}_3(t)| < 2q_0/d_{2v}$, respectively. Similarly, the inequality $|r_1(t)| < p_1$ for (5) and the inequality $|r_3(t)| < q_1$ for (6) yield $|e_2(t)| < p_1/k_{1p}$, $|\dot{e}_2(t)| < 2p_1/k_{1v}$ and $|e_2(t)| < q_1/k_{2p}$, $|\dot{e}_2(t)| < 2q_1/k_{2v}$, respectively. Finally, based on the definitions of (7) and the triangular inequality, the corresponding results are achieved. Q.E.D.

Appendix E
Proof of Theorem 4

Define the following Lyapunov function:

$$V(t) = \left(s_1^2(t) + s_2^2(t)\right)/2 > 0, \qquad \text{as } s_1(t) \neq 0 \text{ or } s_2(t) \neq 0. \tag{E1}$$

Taking the time derivative of (E1), substituting (10) and (11) and (13), (14), (15), and (16) into (E1), and using the fact that $d_{1v}b_{41}(x) > 0$ and $d_{2v}b_{62}(x) > 0\ \forall x(t)$ yields

$$\begin{aligned}
\dot{V}(t) &= s_1(t)\dot{s}_1(t) + s_2(t)\dot{s}_2(t) \\
&= s_1(t)\{d_{1v}\left[\ddot{r}_1(t) - a_4(x) - b_{41}(x)u_1(t) - c_4(x,t)\right] \\
&\qquad + d_{1p}\left(\dot{r}_1(t) - x_4(t) - c_1(x,t)\right)\} \\
&\quad + s_2(t)\{d_{2v}\left[\ddot{r}_3(t) - a_6(x) - b_{62}(x)u_2(t) - c_6(x,t)\right] \\
&\qquad + d_{2p}\left(\dot{r}_3(t) - x_6(t) - c_3(x,t)\right)\} \\
&= s_1(t)h_1(x,t) - d_{1v}b_{41}(x)g_{u_1}s_1^2(t) \\
&\quad - |s_1(t)|\, d_{1v}b_{41}(x)g_{u_1}\Delta_1(t) + s_2(t)h_2(x,t) \\
&\quad - d_{2v}b_{62}(x)g_{u_2}s_2^2(t) - |s_2(t)|\, d_{2v}b_{62}(x)g_{u_2}\Delta_2(t) \\
&\le |s_1(t)|\left\{|h_1(x,t)| - d_{1v}b_{41}(x)g_{u_1}\Delta_1(t)\right\} \\
&\quad + |s_2(t)|\left\{|h_2(x,t)| - d_{2v}b_{62}(x)g_{u_2}\Delta_2(t)\right\} \\
&\le -\lambda\left\{|s_1(t)| + |s_2(t)|\right\} \\
&\le -\lambda\sqrt{s_1^2(t) + s_2^2(t)} \\
&= -\lambda\sqrt{2V(t)}
\end{aligned} \tag{E2}$$

where $\lambda = \min(\lambda_1, \lambda_2)$. Then, the solution of the inequality (E2) for the initial time t_0 and the initial value $s(t_0) = [\, s_1(t_0) \quad s_2(t_0)\,]^{\mathrm{T}}$ is described as follows:

$$t - t_0 \le \|s(t_0)\|/\lambda \tag{E3}$$

where t stands for the time that the operating point hits the sliding surface (i.e., $s(t) = 0$) and $t - t_0$ denotes the finite time to approach the sliding surface. Once the operating point reaches the stable sliding surface (3) and (4), the tracking error asymptotically converges to zero. Q.E.D.

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